

Domain-Specific Evaluation of Text-to-Speech Systems: A Multi-Metric Benchmarking Study

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ABSTRACT

Recent advances in neural text-to-speech (TTS) systems have substantially improved speech naturalness and intelligibility across many languages. However, comprehensive evaluation methodologies that jointly assess perceptual quality, speaker similarity, and acoustic fidelity across diverse speech domains remain limited, particularly for low-resource and underrepresented languages. This paper presents a reproducible, multi-metric benchmarking framework for systematic evaluation of modern TTS systems through domain-specific analysis. The proposed framework integrates complementary subjective and objective evaluation protocols and is demonstrated through a comprehensive case study on a representative low-resource language spanning four speech domains: Formal, Conversational, Literary/Storytelling, and Emotional. Four state-of-the-art TTS systems—Indic-Parler-TTS, MMS-TTS, Microsoft Edge TTS, and Google Gemini TTS—are evaluated using MUSHRA listening tests, ABX discrimination tests, speaker similarity scoring with Resemblyzer, and acoustic analyses based on mel-cepstral distortion (MCD) and F0 RMSE over 960 audio pairs. Results reveal substantial variation in TTS performance across speech domains, with emotional speech consistently presenting the greatest synthesis challenge (mean MCD 12.03 dB; mean F0 RMSE 889 cents), while conversational speech achieves the highest overall acoustic fidelity. Beyond the empirical findings, this work provides a reproducible evaluation framework, publicly releasing evaluation scripts, result tables, and executable Colab notebooks to support standardized benchmarking and future research on TTS evaluation for low-resource languages. All evaluation scripts, result tables, and curated Colab notebooks are *publicly released* at <https://github.com/Shifa402/Urdu-Synthetic-Speech-Evaluation>.

1. Introduction

Text-to-speech (TTS) synthesis has undergone a dramatic transformation over the past decade, with neural end-to-end architectures now producing speech that approaches or, in some perceptual dimensions, exceeds the quality of natural recordings in high-resource languages (Lyth and King, 2024; Pratap, Tjandra, Shi, Tomasello, Babu, Kundu, Elkahky, Ni, Vyas, Fazel-Zarandi, Baevski, Adi, Zhang, Hsu, Conneau and Auli, 2023). Yet progress in *how* synthesis quality is measured has not kept pace with progress in model capability. Across languages, comparative TTS studies still rely disproportionately on single-domain, single-metric protocols that cannot reveal whether a system is fit for a specific communicative purpose.

Even in English and other well-resourced settings, evaluation practice remains narrow. Most benchmark studies report short neutral utterances rated via the Mean Opinion Score (MOS) (Kirkland, Mehta, Lameris, Henter and Szekely, 2023; Le Maguer et al., 2024). As synthetic speech quality improves, MOS scores saturate at the top of the rating scale, obscuring perceptual differences between high-performing systems. More fundamentally, a single-domain evaluation cannot show whether a voice that sounds natural on read-aloud news text will remain acceptable in conversational dialogue, storytelling, or emotionally expressive

speech, dimensions directly relevant to real-world deployment. Recent work therefore advocates *task- and situation-specific* evaluation that assesses whether speech fulfils its intended communicative role rather than limiting comparison to surface-level transmission quality (Bailly et al., 2025).

Urdu provides a timely case study for this broader evaluation problem. Multiple publicly accessible synthesis systems, including MMS-TTS (Pratap et al., 2023), Indic-Parler-TTS (Lyth and King, 2024; Lacombe, Srivastav and Gandhi, 2024), Microsoft Edge TTS, and Google Gemini TTS, are now available, yet no study has compared them systematically across communicative domains. Prior Urdu work has almost exclusively used short neutral prompts and MOS-style ratings (Naseem et al., 2025; Arshad et al., 2022), leaving open the question of whether system rankings are stable when speech style, prosody, and pragmatic context change.

This paper addresses that gap by presenting a domain-stratified benchmark evaluation with Urdu as the target language. Four publicly available systems, namely Indic-Parler-TTS, MMS-TTS, Microsoft Edge TTS (*ur-PK-AsadNeural*), and Google Gemini TTS, are evaluated across four linguistically and pragmatically distinct speech domains: **Formal**, **Conversational**, **Literary/Storytelling**, and **Emotional**. Each domain is sourced from a dedicated dataset and evaluated using a multi-metric pipeline comprising subjective MUSHRA listening tests (International Telecommunication Union, 2015), ABX discrimination tests, speaker-identity similarity scoring via Resemblyzer (Wan, Wang, Papir and Moreno, 2018), mel-cepstral distortion (MCD), and fundamental frequency root-mean-square error (F0 RMSE)

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over 960 audio pairs. The combination of a domain-stratified corpus, multiple complementary metrics, and a fully reproducible pipeline constitutes a methodological advance over prior single-domain, MOS-only TTS evaluations.

The contributions of this paper are as follows:

1. A *domain-stratified benchmarking framework* is introduced for systematic evaluation of TTS systems across diverse speech styles, including Formal, Conversational, Literary/Storytelling, and Emotional domains. The framework is demonstrated through a comprehensive case study involving four TTS systems, three speech datasets, and 960 evaluated audio pairs.
2. A *multi-dimensional evaluation pipeline* is proposed that integrates perceptual, discriminative, acoustic, and speaker-similarity measures, combining MUSHRA-style listening tests, ABX discrimination, MCD, speaker-embedding cosine similarity, and F0 RMSE. This provides a more holistic assessment of synthesis quality than any single evaluation criterion.
3. A *reproducible evaluation resource* is publicly released, including evaluation scripts, result tables, and executable Colab notebooks at <https://github.com/Shifa402/Urdu-Synthetic-Speech-Evaluation>. These resources provide an extensible foundation for future benchmarking of TTS systems across languages, domains, and architectures.

2. Related Work

This section reviews the literature relevant to the evaluation of modern text-to-speech (TTS) systems, advances in speech synthesis for low-resource languages, and prior work on Urdu and South Asian TTS. Together, these studies establish the broader context for this work, highlighting the growing need for comprehensive evaluation methodologies and the limited availability of domain-specific benchmarks for low-resource languages.

2.1. TTS Evaluation Methodology

Subjective evaluation via the Mean Opinion Score (MOS), standardised by ITU-T P.800, has been the dominant evaluation protocol in speech synthesis for decades. MOS instructs listeners to rate each stimulus in isolation on a five-point absolute category rating scale. While its simplicity has made it universally adopted, a growing body of evidence documents its limitations for evaluating modern neural TTS. As the quality of synthetic speech has improved, MOS scores exhibit a *ceiling effect*: high-performing systems cluster at the top of the scale, making it statistically difficult to distinguish between them or between high-quality synthetic speech and natural recordings (Le Maguer et al., 2024). Kirkland et al. (2023) conducted a meta-analysis of Interspeech and SSW papers and found that researchers underreport methodological details, vary widely in scale labels and instructions, and obtain results that are not transferable across studies—meaning MOS scores from different publications cannot be compared without re-running all systems jointly.

The Multiple Stimuli with Hidden Reference and Anchor (MUSHRA) protocol (International Telecommunication Union, 2015) addresses several of these shortcomings. Unlike MOS, MUSHRA presents the hidden reference, one or more anchor conditions, and all systems under test simultaneously, enabling listeners to make direct comparisons on a continuous 0–100 scale. The simultaneous presentation increases sensitivity to inter-system differences and reduces the range effect that distorts absolute MOS ratings (Chiang and Lee, 2024). MUSHRA was originally designed for intermediate-quality audio codecs; however, its relative comparison framework is well-suited to TTS evaluation where the goal is to rank systems against a natural speech reference, and it is increasingly preferred in benchmark studies involving multiple systems (Le Maguer et al., 2024).

Objective metrics complement subjective tests by providing scalable, reproducible measurements that do not require listener recruitment. Mel-cepstral distortion (MCD) (Kubichek, 1993), computed from DTW-aligned mel-cepstral coefficient sequences, quantifies spectral envelope similarity between reference and synthetic speech. Fundamental frequency RMSE (F0 RMSE), expressed in cents on a logarithmic scale (Kominek and Black, 2008), measures pitch accuracy in a perceptually meaningful unit where approximately five cents approximates one just-noticeable difference in pitch. Speaker embedding cosine similarity, computed via the GE2E-trained encoder of Resemblyzer (Wan et al., 2018), measures whether the synthesised voice preserves the identity characteristics of the reference speaker and has been widely adopted in zero-shot TTS evaluation (Wang et al., 2023b). Taken together, MCD and F0 RMSE capture acoustic similarity at the spectral and prosodic levels, while speaker embedding similarity measures a higher-level identity dimension that the two signal-level metrics cannot.

A broader trend in the field is a movement away from aggregate quality metrics toward *task- and situation-specific* evaluation (Bailly et al., 2025). MOS and MUSHRA were designed to measure transmission quality rather than to assess whether speech fulfils its communicative purpose. Evaluation should instead ask whether a voice is appropriate for its intended use case, such as news reading, storytelling, emotional narration, or conversational dialogue, and whether domain-specific prosodic and stylistic requirements are met. The present work operationalises this principle by structuring the evaluation framework around four communicatively distinct domains.

2.2. TTS for Low-Resource Languages

The challenge of building high-quality TTS for under-resourced languages is fundamentally a data problem: expressive, multi-speaker, studio-quality training corpora do not exist at scale for the majority of the world’s languages. Two broad strategies have emerged to address this. The first is *transfer learning*, which exploits acoustic and phonological overlap between high-resource parent languages and the target low-resource language. Approaches that initialise from English Tacotron or FastSpeech models and fine-tune

on small Urdu or South Asian corpora have demonstrated that cross-lingual transfer can yield intelligible synthesis with limited target-language data (Arshad et al., 2022). The second strategy is *massively multilingual pre-training*. Pratap et al. (2023) scaled VITS-based synthesis (Kim, Kong and Son, 2021) to over 1,100 languages by training on aligned religious text recordings, providing a publicly available Urdu model (facebook/mms-tts-urd-script_arabic) that serves as a strong baseline in zero-resource settings. Parallel efforts for Indian languages have explored multi-speaker and multilingual VITS training across thirteen Indo-Aryan and Dravidian languages, finding that shared acoustic models improve both objective metrics and subjective naturalness compared to monolingual systems, though gains are uneven across language families (Jain et al., 2022).

More recently, *description-conditioned language models* have emerged as a paradigm shift in controllable TTS. Parler-TTS (Lyth and King, 2024; Lacombe et al., 2024) encodes a natural-language speaker description through a frozen Flan-T5 encoder (Chung et al., 2022) and generates audio tokens via a cross-attention-conditioned causal Transformer over a Descript Audio Codec with nine residual vector quantisation codebooks. This architecture, adopted in ai4bharat/indic-parler-tts, enables fine-grained voice control without requiring speaker embeddings extracted from reference recordings—a significant advantage in low-resource settings where clean reference recordings for target speakers may be unavailable.

Despite these modelling advances, evaluation practice for low-resource TTS has lagged. Single-metric intelligibility checks such as word or character error rate computed via ASR round-trips remain common but are insufficient: they fail to capture naturalness, prosodic quality, or domain suitability, and are blind to failures in language identification and script fidelity that matter for non-Latin-script languages such as Urdu (Anonymous, 2025). The lack of standardised, multi-domain evaluation benchmarks for South Asian languages means that practitioners have no principled basis for selecting the most appropriate system for a given application.

2.3. Urdu and South Asian Speech Synthesis

Urdu presents a distinct set of challenges for TTS that are not shared by better-resourced languages. Its Nastaliq-style Perso-Arabic script encodes vowels as optional diacritics (*i’raab*), creating text normalisation and grapheme-to-phoneme ambiguities that most multilingual models handle poorly. The language’s phonological proximity to Hindi—while sharing largely the same spoken phoneme inventory—means that models trained on Hindi data can transfer to Urdu synthesis, but the resulting systems often produce unnaturally Hindustani prosody. Additionally, Urdu-English code-switching is prevalent in natural speech, a phenomenon that benchmark evaluations on monolingual read-speech corpora cannot capture.

Early Urdu TTS systems relied on concatenative synthesis or statistical parametric models before the shift to

neural synthesis. Transfer learning from English Tacotron and Transformer-TTS has demonstrated that pre-trained acoustic models can be adapted to Urdu with limited in-language data (Arshad et al., 2022). Naseem et al. (2025) compared Tacotron-based Urdu models against MMS-TTS using MOS, Semantically Unstructured Sentence (SUS) tests, and comprehension evaluations, finding that Tacotron-based models with script-specific text analysis modules outperform MMS in intelligibility and naturalness on read speech, while MMS shows relative advantages on certain word-boundary consonants. Despite this, evaluation in Naseem et al. (2025) is conducted entirely on read-speech prompts with no examination of domain-specific performance variation.

For Indian languages more broadly, Jain et al. (2022) evaluated multi-speaker VITS systems across thirteen languages, reporting MOS and objective metrics but, again, using a single-domain read-speech protocol. The absence of domain-stratified evaluation corpora for South Asian languages reflects a general methodological deficit that extends beyond Urdu: to our knowledge, no published benchmark exists that systematically evaluates how synthesis quality varies across communicative styles for any South Asian language. This paper provides such a benchmark for Urdu, establishing a reproducible evaluation framework that can be extended to related languages as dedicated datasets become available.

3. Methodology for TTS System Selection

This section describes the selection process for the four text-to-speech (TTS) systems evaluated in this study. It first presents the inclusion criteria used to ensure fair and representative system selection, then outlines the architecture and implementation details of each system, and finally compares their key characteristics relevant to the subsequent evaluation.

3.1. Selection Criteria

System selection was governed by three operationally defined criteria applied jointly. First, each system must provide *publicly accessible, programmatic Urdu synthesis* at the time of data collection (November–December 2024), either through a released model checkpoint under an open licence or through a documented cloud API endpoint, such that any research group can reproduce inference without institutional agreements or manual access requests. Second, each system must accept *Urdu Perso-Arabic Unicode text as direct input* without requiring the evaluator to perform manual transliteration or romanisation; imposing such pre-processing would introduce a confound external to the synthesis engine and would not reflect realistic deployment conditions. Third, the selected systems must collectively span *architecturally and methodologically distinct paradigms* currently represented in the TTS literature, specifically, massively multilingual end-to-end synthesis, description-conditioned audio

language modelling, commercial neural TTS, and large audio language model generation, so that benchmark findings generalise beyond any single modelling approach.

Applying these criteria yielded four systems: MMS-TTS (Pratap et al., 2023), Indic-Parler-TTS (Lyth and King, 2024; Lacombe et al., 2024), Microsoft Edge TTS (*ur-PK-AsadNeural*), and Google Gemini TTS. Two candidate systems were considered but excluded. A publicly available Tacotron-2 fine-tune trained on the Urdu Mozilla Common Voice corpus was withdrawn from its Hugging Face repository prior to data collection, precluding reproducible inference. A FastSpeech-2 checkpoint trained on an institutional corpus at the National University of Sciences and Technology (NUST) required institutional VPN access, violating the public accessibility criterion. No other Urdu-capable TTS systems meeting all three criteria were identified in the literature or public model repositories at the time of selection.

3.2. Description of Evaluated TTS Models

MMS-TTS ([facebook/mms-tts-urd-script_arabic](https://github.com/facebook/mms-tts-urd-script_arabic)). Meta’s Massively Multilingual Speech (MMS) project extended VITS-based end-to-end TTS (Kim et al., 2021) to 1,107 languages by training on sentence-aligned read-speech recordings sourced from publicly available multilingual corpora (Pratap et al., 2023). VITS jointly optimises a conditional variational autoencoder over mel-spectrogram frames, a normalising flow-based prior, and an adversarial waveform decoder, enabling single-stage, non-autoregressive synthesis (Kim et al., 2021). Text is processed through the UROMAN universal grapheme-to-phoneme converter before being passed to the acoustic model (Pratap et al., 2023). The Urdu checkpoint targets the Perso-Arabic script partition and, per the MMS training documentation, is derived predominantly from a single-speaker read-speech source. Inference was performed locally via the Hugging Face transformers library (v4.40.0) at a native output sample rate of 16 kHz. MMS-TTS is the only fully open-weight, locally executable system in this evaluation, which has practical implications for deployment in resource-constrained or low-connectivity environments.

Indic-Parler-TTS ([ai4bharat / indic-parler-tts](https://github.com/ai4bharat/indic-parler-tts)). Parler-TTS (Lyth and King, 2024; Lacombe et al., 2024) is a description-conditioned audio language model that generates speech by conditioning a causal Transformer decoder on both the target text and a natural-language speaker description. The description is encoded by a frozen Flan-T5-Large encoder (Chung et al., 2022), and the resulting representations cross-attend to a GPT-2-scale decoder that predicts discrete audio tokens from a nine-codebook Descript Audio Codec (DAC) (Lacombe et al., 2024). A corresponding DAC decoder renders waveforms at 44.1 kHz. The [ai4bharat/indic-parler-tts](https://github.com/ai4bharat/indic-parler-tts) checkpoint (AI4Bharat, 2024) extends this architecture to twenty-two Indic languages, including Urdu, through continued pre-training on a multilingual read-speech and audiobook corpus assembled by AI4Bharat. Because voice

characteristics are specified through free-text descriptions rather than extracted speaker embeddings, the model does not require a reference recording to condition synthesis, a property that is advantageous in low-resource settings where clean reference audio for target speakers may be unavailable (Lacombe et al., 2024). A fixed description specifying a male speaker with a clear, moderate-paced voice was used uniformly across all evaluation domains to control for inter-domain description variation.

Microsoft Edge TTS (*ur-PK-AsadNeural*). Microsoft Edge TTS is a cloud-based neural TTS service that exposes synthesis capability through a documented WebSocket endpoint accessed via the open-source *edge-tts* Python library (v6.1.9). The *ur-PK-AsadNeural* voice targets Pakistani Urdu and is part of Microsoft’s Azure Cognitive Services Neural TTS platform (Microsoft Corporation, 2023), which employs a Transformer-based acoustic model combined with a neural vocoder, trained on professionally recorded studio speech. Synthesis is produced at 24 kHz. Microsoft does not publicly disclose training corpus composition, model architecture specifications, or speaker recording details for production neural voices, and accordingly this system is treated as an opaque commercial baseline within the evaluation. Its inclusion is motivated by the documented adoption of the *AsadNeural* voice in Urdu-language assistive technology and digital media applications, making its performance profile directly relevant to practitioners evaluating deployment options.

Google Gemini TTS. Google Gemini TTS generates speech natively through the Gemini 2.5 Flash model with audio output enabled, accessed via the *google-generativeai* Python SDK (v0.8.3) using the *gemini-2.5-flash-preview-tts* endpoint. Unlike conventional cascaded TTS pipelines that decompose synthesis into independent text analysis, acoustic modelling, and vocoding stages, Gemini TTS renders speech directly from the language model’s contextual representations through a dedicated speech decoder, a design that has been reported to improve prosodic coherence over multi-sentence inputs (Google DeepMind, 2025). The system is configured using the *PrebuiltVoiceConfig* parameter; the *Charon* voice preset was applied uniformly across all evaluation domains to isolate domain-specific synthesis variation from speaker-selection effects. Output audio was produced at 24 kHz. Google does not publicly disclose the Urdu training data volume, language-specific fine-tuning procedures, or internal model specifications for this system. To the best of our knowledge, this paper presents the first published objective and subjective evaluation of Gemini TTS on Urdu speech.

3.3. Comparative System Properties

Table 1 summarises the architectural, training, and deployment properties of the four systems. Differences along four dimensions are particularly relevant to interpreting domain-specific results. (i) *Synthesis paradigm*: MMS-TTS uses non-autoregressive flow-based generation; Indic-Parler-TTS uses autoregressive discrete token prediction; Edge TTS

Table 1

Architectural and deployment properties of the four evaluated Urdu TTS systems. Proprietary systems are marked with (†) to indicate that architecture and training details are not publicly disclosed.

System	Paradigm	Training Scope	Speaker Control	Rate	Access
MMS-TTS	VITS: VAE + flow + GAN (Kim et al., 2021)	Massively multilingual (1,107 lang.)	Single speaker (M)	16 kHz	Open weight
Indic-Parler-TTS	Autoregressive LM + DAC (Lacombe et al., 2024)	22 Indic languages	Description-conditioned	44.1 kHz	Open weight
Edge TTS†	Neural (undisclosed) (Microsoft Corporation, 2023)	Proprietary (ur-PK)	Preset (M, ur-PK)	24 kHz	Cloud API
Gemini TTS†	Audio LM, direct render (Google DeepMind, 2025)	Proprietary multilingual	Preset voices	24 kHz	Cloud API

uses an undisclosed Transformer-based pipeline; and Gemini TTS uses direct audio rendering from a large language model. (ii) *Training scope*: MMS-TTS and Gemini TTS are trained at massive multilingual scale, whereas Indic-Parler-TTS targets a defined set of twenty-two Indic languages and Edge TTS is a dedicated Pakistani Urdu voice. (iii) *Speaker control*: MMS-TTS produces a fixed single-speaker output; Indic-Parler-TTS accepts free-text description conditioning; Edge and Gemini TTS offer discrete preset selection. (iv) *Output sample rate*: 16 kHz (MMS-TTS), 24 kHz (Edge TTS, Gemini TTS), and 44.1 kHz (Indic-Parler-TTS); prior to objective evaluation, audio was resampled at metric-specific rates (22.05 kHz for MCD and 16 kHz for F0 RMSE and Resemblyzer speaker embeddings) to match each feature extractor’s requirements (see Section 5).

4. Domain-Wise Dataset Setup

This section describes the construction and organisation of the evaluation corpus used in this study. It motivates the need for domain diversity in TTS evaluation and presents each of the four speech domains, namely Formal, Conversational, Emotional, and Literary/Storytelling.

4.1. Need for Domain Diversity in TTS Evaluation

The construction of a domain-diverse dataset is an important factor in evaluating modern text-to-speech (TTS) systems, particularly for low-resource languages such as Urdu. Recent research has shown that the quality of speech synthesis is strongly influenced not only by model architecture but also by the diversity and distribution of evaluation data between speaking styles and domains (Kong, Kim and Bae, 2020; Tan, Qin, Soong and Liu, 2021). Traditional evaluation datasets often focus on homogeneous read speech, which limits their ability to capture real-world variability in prosody, emotion, and communicative intent. As TTS systems continue to evolve towards more expressive and controllable generation, the need for diverse and domain-aware datasets has become increasingly important for robust benchmarking.

In this study, the evaluation dataset is organized into four distinct speech domains, namely Formal, Conversational, Literary/Storytelling, and Emotional. Each domain

represents a different communicative and stylistic setting, characterized by unique linguistic and prosodic patterns. This domain selection aligns with recent advances in expressive and controllable speech synthesis, where systems are expected to generalize across multiple speaking styles rather than in a single uniform voice setting (Wang et al., 2017; Kim et al., 2021). The resulting dataset design enables a more comprehensive evaluation of Urdu TTS systems under varied real-world conditions, supporting fine-grained analysis across both neutral and expressive speech scenarios.

4.2. Domain 1: Formal Speech

This subsection defines the Formal speech domain and describes the datasets selected to represent it in the evaluation corpus. Together, these datasets capture the linguistic and communicative characteristics of formal speech required for domain-specific evaluation.

4.2.1. Domain Characteristics

The Formal domain represents controlled and structured speech produced in situations where the primary objective is clarity, correctness, and linguistic precision. This can include read-aloud scenarios such as news broadcasting, academic narration, and scripted announcements, where speakers typically follow prepared text without spontaneous deviations. The importance of this domain lies in its role as a baseline condition for text-to-speech evaluation, as it minimizes variability in speaking style, emotion, and discourse structure.

In the context of TTS systems, formal read speech provides a stable reference point for assessing fundamental synthesis capabilities such as pronunciation accuracy, fluency, and prosodic consistency. Since the linguistic input in this domain is well-formed and predictable, it allows for a more controlled evaluation of system performance, making it widely used in both benchmarking and comparative analysis of speech synthesis models.

4.2.2. Selected Dataset(s)

This domain is constructed using publicly available multilingual speech corpora that provide high-quality, studio-like read speech recordings under controlled acoustic conditions. The Google FLEURS dataset is used as a primary

source due to its well-standardized recording protocol, multilingual coverage, and consistent speech delivery across speakers and languages. In this study, only Urdu-language utterances are selected to ensure linguistic relevance, while speaker-level filtering is applied to retain male voice recordings for maintaining uniformity across all experimental conditions.

Dataset link: <https://huggingface.co/datasets/google/fleurs>

In addition, the UrduSpeech dataset is incorporated as a complementary source of formal read speech. This dataset provides high-quality Urdu speech with multiple speaking styles and linguistic variations. For the purpose of this work, only Urdu samples are retained, as the corpus also contains closely related Indic language data such as Kashmiri. Furthermore, style-based filtering is applied to select Wiki-style utterances, ensuring adherence to a formal and structured reading modality. Gender-based filtering is again applied to retain male speakers, ensuring consistency across datasets used in the evaluation framework.

Dataset link: <https://huggingface.co/datasets/humairawan/UrduSpeech>

4.3. Domain 2: Conversational Speech

This subsection introduces the Conversational speech domain and the dataset selected to represent it within the evaluation corpus.

4.3.1. Domain Characteristics

The conversational domain represents spontaneous and semi-structured dialogue-based communication that occurs in everyday interactions. Unlike formal read speech, this domain is characterized by natural speaking patterns, including variations in intonation, informal phrasing, and contextual dependencies between utterances. The importance of this domain lies in its ability to capture realistic speech behavior, making it essential for evaluating the robustness of text-to-speech systems in non-scripted and socially interactive scenarios. This domain introduces greater variability in prosody, rhythm, and speaking style, providing a more challenging evaluation setting compared to controlled speech conditions.

4.3.2. Selected Dataset(s)

The data for this domain are derived from the UrduSpeech dataset publicly available on Hugging Face. Since the dataset contains multilingual recordings, only Urdu-language utterances are retained for this study. A style-based filtering criterion is applied to isolate conversational speech samples and only recordings labeled as conversational are included. With both male and female voice audios being present, only male speaker recordings are selected.

Dataset link: <https://huggingface.co/datasets/humairawan/UrduSpeech>

4.4. Domain 3: Emotional Speech

This subsection describes the Emotional speech domain, outlining the acoustic and affective properties that distinguish it from other domains and the dataset from which emotional utterances are drawn.

4.4.1. Domain Characteristics

The emotional domain represents expressive speech characterized by systematic variations in prosody, pitch, intensity, and speaking rate that correspond to different affective states. Unlike formal or conversational speech, this domain explicitly encodes emotional information within the acoustic signal, making it significantly more variable and challenging for text-to-speech systems. The inclusion of emotional speech is essential for evaluating expressive speech synthesis capabilities, as it tests whether a system can generate natural and contextually appropriate emotional variation beyond neutral speaking styles.

4.4.2. Selected Dataset(s)

The emotional speech data is obtained from the Urdu Speech Emotion Corpus (UrSEC), a comprehensive dataset designed for speech emotion recognition in the Urdu language. The dataset contains 3,500 annotated speech samples collected from professional actors and covers seven emotional categories: Angry, Fear, Boredom, Disgust, Happy, Neutral, and Sad. The recordings are sourced from naturally spoken Urdu dialogues derived from drama serials and telefilms, ensuring realistic and contextually rich utterances.

Dataset link: <https://data.mendeley.com/datasets/jcpfjnk5c2/4>

For the purpose of this study, a balanced subset of samples is selected across all emotion categories to ensure uniform representation of affective states. The Neutral category is excluded from evaluation to avoid overlap with the Formal and Conversational Speech domain defined in this work. Only male-speaker recordings are retained to maintain consistency across all experimental domains.

4.5. Domain 4: Literary/Storytelling Speech

This subsection describes the Literary/Storytelling speech domain, detailing the narrative and prosodic features that characterise it and the dataset used for its construction.

4.5.1. Domain Characteristics

This domain represents expressive narrative speech used in contexts such as storytelling, book reading, and dramatic narration. Unlike formal or conversational speech, this domain is characterized by dynamic prosodic variation, extended utterance structure, and deliberate modulation of pitch and rhythm to convey narrative meaning and emotional context. This makes it an important evaluation setting for text-to-speech systems aimed at expressive synthesis, as it tests the ability of models to generate natural and engaging narrative speech beyond neutral or conversational styles.

4.5.2. Selected Dataset(s)

The Literary/Storytelling domain is constructed using the diverse UrduSpeech dataset, which contains multiple speaking styles. From this dataset, only Urdu-language samples are selected. Additionally, style-based filtering is applied to retain BOOK-style recordings, which best represent storytelling and literary narration. Speaker-level filtering is further applied to maintain male-speaker consistency across all evaluation domains.

Dataset link: <https://huggingface.co/datasets/humairawan/UrduSpeech>

5. Evaluation Methodology

This section presents the evaluation framework employed in this study. To provide a comprehensive assessment of text-to-speech (TTS) quality, the framework integrates complementary subjective and objective evaluation methodologies, capturing perceptual quality, speaker similarity, and acoustic fidelity. Together, these evaluation components provide a holistic basis for comparing TTS systems across different speech domains while ensuring a reproducible and consistent benchmarking process.

5.1. Evaluation Framework Design

The evaluation pipeline is organised around two complementary design principles. The first is *metric complementarity*: no single metric captures all perceptually relevant dimensions of synthesis quality (Kirkland et al., 2023; Le Maguer et al., 2024), and accordingly the framework combines subjective listening tests with three classes of objective measurement, namely speaker identity preservation, spectral envelope fidelity, and prosodic accuracy. The second is *domain stratification*: all metrics are computed separately for each of the four speech domains (Formal, Conversational, Literary/Storytelling, and Emotional) rather than aggregated across domains, so that domain-specific performance variation is directly observable rather than obscured by averaging. Together, these principles operationalise the task-and-situation-specific evaluation paradigm advocated by Bailly et al. (2025) for the specific case of Urdu TTS.

The complete pipeline proceeds as follows. For each domain, reference recordings are paired with synthetic outputs from each of the four TTS systems at the utterance level, yielding 960 audio pairs in total (4 systems $\times$ 4 domains $\times$ 60 utterances per domain). Each pair is then passed through four evaluation modules: a subjective MUSHRA-style listening test, a speaker similarity scoring procedure via Resemblyzer (Wan et al., 2018), mel-cepstral distortion (MCD) computation (Kubichek, 1993), and fundamental frequency root-mean-square error (F0 RMSE) computation (Kominek and Black, 2008). An additional forced-choice ABX discrimination test is conducted as a second-stage evaluation, applied to the two top-performing systems identified through the MUSHRA stage, to assess perceptual indistinguishability from natural speech. All evaluation scripts and result tables are publicly available at <https://github.com/Shifa402/Urdu-Synthetic-Speech-Evaluation>.

5.2. Subjective Evaluation

Two complementary subjective evaluation procedures are employed. A MUSHRA-style listening test provides comparative quality ratings across all four systems, while an ABX discrimination test assesses perceptual indistinguishability from natural speech for the two highest-ranked systems.

5.2.1. MUSHRA-Style Listening Test

Subjective evaluation was conducted using a MUSHRA-inspired protocol (International Telecommunication Union, 2015) adapted for online administration via Google Forms. The adaptation retains the core structural elements of MUSHRA, specifically the simultaneous presentation of a hidden reference, a degraded anchor, and all systems under test, while replacing the standard MUSHRA software interface with a structured questionnaire format accessible to participants without specialist software installation. This approach is consistent with practices reported in recent remote TTS evaluation studies (Chiang et al., 2024).

Each evaluation page presented listeners with six audio samples for a given utterance: one reference recording drawn from the source dataset representing original human speech, one anchor recording consisting of a lower-quality degraded version of the same utterance, and four synthetic outputs produced by MMS-TTS, Indic-Parler-TTS, Edge TTS, and Gemini TTS respectively. Listeners were instructed to rate each of the four synthetic outputs on a discrete 1–5 scale across three perceptual dimensions: *Naturalness*, defined as whether the speech sounds human-like and fluent; *Clarity*, defined as whether the speech is intelligible and free of artefacts; and *Overall Speech Quality*, defined as a holistic judgement relative to the reference. The reference and anchor were presented for orientation purposes and were not themselves rated.

A total of 20 native Urdu-speaking listeners participated in the evaluation. Participants were native Urdu speakers within the same demographic range. As the research study was situated in Lahore and its population comprises predominantly native speakers of Urdu, formal language proficiency screening was not considered necessary. The online Google Forms platform was selected because it was accessible to this population without requiring attendance at a dedicated laboratory session, thereby reducing recruitment barriers while maintaining a controlled listening protocol through the standardised instructions detailed above. Listeners were assigned to two domains each, yielding 10 listeners per domain pair, in order to limit session length to a manageable duration and to reduce listener fatigue, which is a recognised source of rating variance in extended subjective speech evaluations (Le Maguer et al., 2024). No formal phonetics training was required; participants were screened for self-reported normal hearing. Each listener evaluated all four systems within their assigned domains, ensuring that within-domain system comparisons are fully within-subject.

It is acknowledged that a panel of 20 listeners, with 10 per domain, is smaller than the listener pools recommended by the ITU-R BS.1534 MUSHRA standard (International Telecommunication Union, 2015), and that the Google Forms interface does not implement the continuous 0–100 slider specified by the standard. These deviations are documented explicitly so that readers may calibrate the strength of the subjective findings accordingly. The objective metrics reported in Section 5.3 do not share these limitations and are accorded commensurate weight in the cross-metric synthesis presented in Section 7.6.

5.2.2. ABX Discrimination Test

The ABX discrimination test was conducted as a second-stage evaluation, applied selectively to the two systems that achieved the highest MUSHRA-style ratings in the first stage, namely Edge TTS and Gemini TTS. This two-stage design is motivated by the observation that ABX discrimination is most informative when applied to high-performing systems, given that lower-quality systems are trivially distinguishable from natural speech and contribute limited diagnostic value to a forced-choice paradigm (Schatz, Peddinti, Bach, Jansen, Hermansky and Dupoux, 2013). By restricting the ABX stage to the top-ranked systems identified through the MUSHRA procedure, the test directly addresses the more challenging and practically relevant question of whether the best-performing Urdu TTS systems approach perceptual indistinguishability from natural speech.

In each ABX trial, listeners were presented with three stimuli: a reference utterance (A), a second presentation of the same reference (B), and an unknown sample (X) drawn from either the reference or a synthetic output produced by Edge-TTS or Gemini TTS. Listeners indicated whether X was perceptually more similar to A or to B. A discrimination rate significantly above 50% indicates that the synthetic output remains perceptually distinguishable from natural speech, whereas a rate at or near 50% indicates near-indistinguishability. The test was administered under conditions identical to those of the MUSHRA-style listening test, via Google Forms, using the same standardised listening instructions, and within the same online session. Participants were native Urdu speakers aged 18–25 years, recruited from the immediate social and academic networks of the research team. The listener panel was gender-balanced, with an equal number of male and female participants, to mitigate potential systematic gender-based perceptual bias in the discrimination of synthesised Urdu speech. As with the MUSHRA-style evaluation, listeners were assigned to two domains each, yielding 10 listeners per domain pair, and both systems were evaluated within each listener’s assigned domains to preserve the within-subject comparison structure.

5.3. Objective Evaluation

Three complementary objective metrics are employed to assess different aspects of synthesis quality. Speaker similarity, computed using Resemblyzer, evaluates voice identity

preservation; mel-cepstral distortion (MCD) measures spectral envelope fidelity; and fundamental frequency root mean square error (F0 RMSE) quantifies pitch contour accuracy.

5.3.1. Speaker Similarity via Resemblyzer

Speaker identity preservation was quantified using the cosine similarity between d-vector speaker embeddings extracted by Resemblyzer (Wan et al., 2018). Resemblyzer implements the Generalised End-to-End (GE2E) loss-trained LSTM encoder of Wan et al. (2018), which maps a variable-length speech segment to a fixed-dimensional embedding in a speaker-discriminative space. For each reference-synthetic pair, embeddings were extracted independently and their cosine similarity was computed; values approaching 1.0 indicate higher speaker identity similarity between the reference and the synthesised output.

One methodological consideration specific to this evaluation concerns MMS-TTS, which produces a fixed single-speaker output whose voice identity is independent of the reference speaker. All reference and synthetic audio in this study is restricted to male speakers, so every comparison is within-gender; nonetheless, the fixed MMS-TTS voice differs in speaker identity from each domain’s reference speaker, and its cosine similarity therefore primarily reflects speaker-identity mismatch rather than synthesis degradation, consistent with cautions raised for speaker-similarity evaluation of single-speaker systems (Wang, Chen, Wu, Zhang, Zhou, Liu, Chen, Liu, Wang, Li et al., 2023a). All audio was resampled to 16 kHz prior to embedding extraction, as required by the Resemblyzer encoder input specification.

5.3.2. Mel-Cepstral Distortion

Mel-cepstral distortion (MCD) (Kubichek, 1993) quantifies spectral envelope similarity between a reference and a synthetic utterance by computing the Euclidean distance between their mel-cepstral coefficient sequences after dynamic time warping (DTW) alignment. Lower MCD values indicate greater spectral similarity. MCD is computed as:

$$\text{MCD} = \frac{10\sqrt{2}}{\ln 10} \frac{1}{T} \sum_{t=1}^T \sqrt{\sum_{k=1}^K (c_{k,t}^{\text{ref}} - c_{k,t}^{\text{syn}})^2} \quad (1)$$

where $c_{k,t}^{\text{ref}}$ and $c_{k,t}^{\text{syn}}$ are the k -th mel-cepstral coefficients of the t -th DTW-aligned reference and synthetic frame pair, T is the number of aligned frames, and $K = 13$ mel-cepstral coefficients are retained following standard practice (Kominek and Black, 2008); the leading scaling constant $10\sqrt{2}/\ln 10 \approx 6.142$ expresses the distortion in decibels. Mel-cepstra were obtained from a WORLD spectral envelope followed by SPTK mel-cepstral analysis (order 13, warping coefficient $\alpha = 0.65$) using the `pymcd` implementation, with DTW alignment on the C_1 – C_{13} coefficients performed by `fastdtw` under a Euclidean cost. Audio was resampled to 22.05 kHz for this analysis, ensuring comparability across systems with differing native output sample rates.

Table 2

Summary of evaluation metrics, the quality dimension each captures, and result directionality.

Metric	Type	Dimension	Better
MUSHRA-style rating	Subjective	Naturalness, clarity, overall quality	Higher
ABX discrimination	Subjective	Perceptual distinguishability from natural speech	Lower
Speaker similarity	Objective	Speaker identity preservation	Higher
MCD	Objective	Spectral envelope fidelity	Lower
F0 RMSE	Objective	Pitch contour accuracy	Lower

5.3.3. Fundamental Frequency RMSE

Pitch accuracy was assessed using the root-mean-square error between the fundamental frequency (F0) contours of the reference and synthetic utterances, expressed in cents on a logarithmic scale (Kominek and Black, 2008):

$$\text{F0 RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N \left(1200 \log_2 \frac{F0_i^{\text{syn}}}{F0_i^{\text{ref}}} \right)^2} \quad (2)$$

where N is the number of voiced frames present in both the reference and synthetic signals after DTW alignment. The cent scale is employed because approximately 5 cents approximates one just-noticeable difference in pitch under typical listening conditions (Kominek and Black, 2008). F0 was extracted with pyworld using the harvest algorithm followed by stonemask refinement (50–800 Hz search range) at a 16 kHz sampling rate, and the reference and synthetic contours were compared on the frame pairs of a DTW alignment. Unvoiced frames and frames in which either the reference or synthetic F0 estimate was detected as zero were excluded from the RMSE computation to avoid artefacts attributable to voicing detection errors. Because each synthetic utterance is compared against a natural reference produced by a different speaker, the reported F0 RMSE reflects both pitch-contour deviation and a speaker-dependent register offset; the interpretation of such reference-distance metrics is discussed further in Section 8.3.

5.4. Evaluation Pipeline Summary

Table 2 summarises the four evaluation modules, the dimension of synthesis quality each addresses, and the directionality of each metric. The complete pipeline, including preprocessing scripts, metric computation notebooks, and raw result tables, is publicly released at <https://github.com/Shifa402/Urdu-Synthetic-Speech-Evaluation> to support reproducibility and to serve as an extensible baseline for future Urdu TTS benchmarking.

6. Experimental Setup

This section presents the implementation details of the synthesis and evaluation pipeline. It describes the inference configuration for each TTS system and the hardware and software environment used to conduct all experiments,

ensuring reproducibility and consistent evaluation across systems.

6.1. Implementation Details

All synthesis and evaluation were implemented in Python within cloud-hosted notebooks. For each domain, the reference utterances were drawn from the source datasets described in Section 4 and exported as single-channel WAV files. Each reference utterance was then synthesised once by every one of the four TTS systems directly from its original Urdu Perso-Arabic transcript, without manual transliteration or romanisation. For every system the speaker or voice configuration was held fixed across all four domains, so that the communicative domain remained the only manipulated factor.

MMS-TTS. The model was instantiated as a Vits-Model with its associated AutoTokenizer from the facebook/mms-tts-urd-script_arabic checkpoint through the Hugging Face transformers library. Waveforms were produced in a single non-autoregressive forward pass under torch.no_grad() at the model’s native 16 kHz output rate. Synthesis is deterministic and no decoding hyperparameters were tuned.

Indic-Parler-TTS. The model was loaded as ParlerTTSForConditionalGeneration from ai4bharat/indic-parler-tts, with separate tokenisers for the transcript and for the natural-language speaker description. A single fixed description, “*Rohit’s voice is clear and natural with a moderate speed and pitch*”, which selects the named male speaker *Rohit*, was used unchanged across all domains to hold voice identity constant. Audio tokens were generated with do_sample=True and temperature = 0.8 and decoded to waveforms at the native 44.1 kHz.

Microsoft Edge TTS. Speech was synthesised through the edge-tts client using the *ur-PK-AsadNeural* voice with the service’s default prosody (rate, pitch, and volume left at their default values). Audio was returned at 24 kHz.

Google Gemini TTS. Speech was generated through the gemini-2.5-flash-preview-tts model with response_modalities set to audio and a PrebuiltVoiceConfig specifying the *Charon* preset voice, again held constant across domains. Audio was returned as 24 kHz PCM.

Across the four domains, 60 reference utterances per domain were used for the objective metrics; for the Formal domain these comprised 30 FLEURS and 30 UrduSpeech Wikipedia-style utterances. This yields 4 systems ×

Table 3

Key software components used for synthesis and objective evaluation.

Component	Role	Version
transformers	MMS-TTS / tokenisers	4.40.0
parler-tts	Indic-Parler-TTS	release at access
edge-tts	Edge TTS client	6.1.9
google-generativeai	Gemini TTS client	0.8.3
resemblyzer	Speaker embeddings	0.1.4
librosa	Audio I/O, resampling	0.11.0
pyworld	F0 extraction (WORLD)	0.3.5
pymcd	Mel-cepstral distortion	0.2.1
pysptk	Mel-cepstral analysis	1.0.1
fastdtw	DTW alignment	0.3.4

4 domains $\times$ 60 = 960 reference–synthetic pairs for speaker similarity, MCD, and F0 RMSE. The subjective listening tests used a 30-utterance subset per domain to limit session length and listener fatigue. Reference and synthetic clips were paired by filename stem. Because the two cloud services (Edge, Gemini) and the sampling-based Indic-Parler-TTS decoder are not seed-controllable, a single synthesis was generated per utterance; MMS-TTS and the two cloud presets are otherwise deterministic for a fixed input.

6.2. Hardware and Software Configuration

The neural synthesis systems (MMS-TTS and Indic-Parler-TTS) and the speaker-embedding extraction were executed on a single CUDA-enabled GPU within a Google Colab environment running Python 3.12, while the two cloud systems (Edge TTS and Gemini TTS) were accessed over their respective network APIs. The mel-cepstral distortion and fundamental-frequency computations are CPU-bound and were run locally.

Speaker similarity was computed with the GE2E-trained VoiceEncoder of Resemblyzer (Wan et al., 2018): reference and synthetic clips were passed through preprocess_wav (16 kHz resampling with voice-activity trimming) and compared by the cosine similarity of their utterance embeddings. The MCD and F0 RMSE were computed by a single in-house script. For MCD, mel-cepstral coefficients (order 13, $\alpha = 0.65$) were extracted from a WORLD spectral envelope via SPTK using the pymcd library at 22.05 kHz, DTW-aligned with fastdtw under a Euclidean cost, and scored with the standard Kominek–Black scaling constant. For F0 RMSE, audio was resampled to 16 kHz mono and F0 was extracted with pyworld (harvest+stonemask, 50–800 Hz), with the reference and synthetic contours compared in cents on DTW frame pairs. Key software components and versions are listed in Table 3.

7. Results

This section presents evaluation results organised by metric, proceeding from subjective perceptual assessment to objective acoustic and speaker identity measures, and

concluding with a cross-metric synthesis. For each metric, results are reported first at the domain level, aggregated across all four systems, and then at the system level within each domain. This organisation directly addresses the core research question: whether synthesis quality varies systematically across communicative domains and whether domain sensitivity is consistent across evaluation dimensions. All 960 reference–synthetic pairs were evaluated without computation failure, confirming complete corpus coverage.

7.1. MUSHRA-Style Listening Test Scores

MUSHRA-style results are analysed at both the domain and system levels. Domain-level results are first presented by aggregating scores across all systems, followed by system-level analyses within each speech domain.

7.1.1. Domain-Level MUSHRA Results

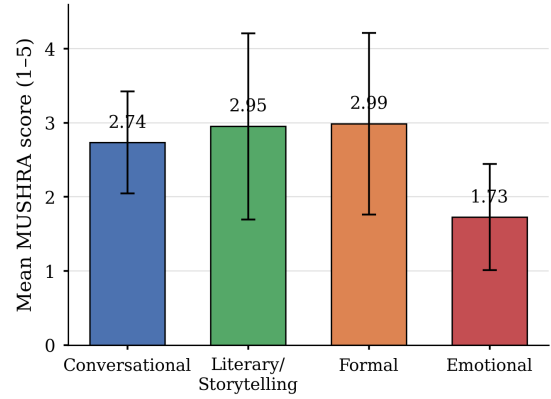


Figure 1: Domain-level mean MUSHRA-style scores.

7.1.2. System-Level MUSHRA Results

Table 4: Per-system mean MUSHRA-style scores (1–5 scale) by domain. Conv. = Conversational, Lit. = Literary/Storytelling, Form. = Formal, Emot. = Emotional. Bold = best system per domain (column).

System	Conv.	Lit.	Form.	Emot.	Overall
Edge TTS	3.093	3.361	3.373	1.963	2.947
Gemini TTS	3.148	3.602	3.244	2.167	3.040
Indic-Parler-TTS	2.148	2.214	2.738	1.296	2.099
MMS-TTS	2.556	2.631	2.619	1.481	2.322

Note: SDs (in parentheses, per domain-system cell) – Edge TTS: 0.442, 1.071, 1.210, 0.666; Gemini TTS: 0.487, 1.097, 1.416, 0.739; Indic-Parler-TTS: 0.650, 1.103, 1.001, 0.496; MMS-TTS: 0.598, 1.232, 1.090, 0.569 (order: Conv., Lit., Form., Emot.).

Gemini TTS remains the top-ranked system overall (3.040), narrowly ahead of Edge TTS (2.947), with Indic-Parler-TTS (2.099) and MMS-TTS (2.322) trailing substantially. Gemini TTS leads on three of four domains (Conversational, Literary/Storytelling, Emotional); Edge TTS leads only on Formal (3.373 vs. 3.244). The Emotional domain again produces the lowest scores for every system, with even the best performer (Gemini, 2.167) falling well below the scale midpoint.

7.2. ABX Discrimination Results

Listeners correctly identified the reference-matching system in 90.7% of trials overall (98/108), well above the 50% chance level in both domains. Accuracy was identical to one decimal place across Literary/Storytelling and Formal (49/54 each), indicating MMS-TTS and Gemini-TTS remain reliably distinguishable from the reference regardless of domain.

Table 5: ABX discrimination accuracy (%), Edge-TTS vs. Gemini-TTS. Chance level is 50%. $n = 54$ trials per domain (9 listeners $\times$ 6 samples).

Domain	Accuracy (%)	n
Literary/Storytelling	90.7	54
Formal	90.7	54
Overall	90.7	108

7.3. Speaker Similarity via Resemblyzer

Speaker similarity results are analysed at the domain, system, and cross-domain levels. Domain-level analyses characterise overall voice identity preservation across speech domains, system-level analyses compare the performance of individual TTS systems, and cross-domain analyses assess the stability of system rankings across different domains.

7.3.1. Domain-Level Speaker Similarity

Table 6 reports mean cosine similarity between reference and synthetic speaker embeddings aggregated across all four systems per domain. A consistent domain-level gradient is observed, with Literary/Storytelling achieving the highest mean similarity (0.6903) and the Emotional domain the lowest (0.5437), representing a difference of 0.1466 cosine similarity units across the domain range.

Table 6: Domain-level mean Resemblyzer cosine similarity aggregated across all four TTS systems ($n = 240$ pairs per domain).

Domain	Mean	SD
Literary/Storytelling	0.6903	0.0793
Conversational	0.6592	0.0739
Formal	0.5758	0.0765
Emotional	0.5437	0.0667

The domain ranking on speaker similarity differs from the domain rankings on MCD and F0 RMSE in one substantive respect: Literary/Storytelling ranks first on speaker similarity (0.6903) but second on MCD (7.87 dB) and jointly first on F0 RMSE (521.24 cents). Conversational ranks second on speaker similarity (0.6592) but first on MCD. This dissociation indicates that speaker identity preservation and spectral envelope fidelity are partially independent dimensions of synthesis quality and that a system may preserve the perceptual characteristics of a speaker’s voice while still diverging acoustically from the reference signal at the frame level. The Formal domain ranks third on speaker similarity (0.5758) despite containing what might be expected to be the most predictable prosodic content, a pattern attributable to within-domain heterogeneity between its two source subsets, as elaborated

in Section 7.3.2. The Emotional domain ranks last on speaker similarity (0.5437), consistent with its ranking on both MCD and F0 RMSE, confirming that emotional speech presents the most challenging conditions for synthesis quality preservation across all three objective dimensions.

7.3.2. System-Level Speaker Similarity

Table 7 reports cosine similarity per system per domain. All reference and synthetic audio is male-speaker. MMS-TTS produces a fixed single-speaker (male) output; because its voice identity is fixed and differs from each domain’s reference speaker, its similarity scores primarily reflect speaker-identity mismatch rather than synthesis quality, and should be interpreted accordingly. Table 8 reports the cross-system emotion-level breakdown.

Boredom yields the lowest cross-system mean similarity (0.5283), a counterintuitive finding given that boredom is a low-arousal state with reduced prosodic variation. A plausible explanation is that boredom in Urdu speech is characterised by atypical pitch declination patterns and reduced loudness that deviate from neutral read-speech baselines in ways distinct from high-arousal emotions, making speaker embedding extraction less reliable even when the acoustic signal is relatively flat. Edge TTS is the only system that achieves its highest emotion-level similarity on Happy (0.6106), while all other systems rank Happy fourth or lower, suggesting that Edge TTS’s neural vocoder better preserves speaker identity under conditions of increased breathiness and higher pitch that characterise happy speech in Urdu.

7.3.3. Cross-Domain Speaker Similarity Pattern

Indic-Parler-TTS leads on Conversational and Literary but ranks last on Formal. MMS-TTS leads on Formal and Emotional but ranks last on Conversational and Literary. This cross-domain rank instability on speaker similarity mirrors the rank reversal observed on MCD in the Emotional domain (Section 7.4) and provides convergent evidence that system selection based on single-domain evaluation would systematically misrepresent deployment performance across communicative contexts. The practical implication is that no currently available Urdu TTS system provides uniformly high speaker identity preservation across the full range of speech domains evaluated in this study.

7.4. Mel-Cepstral Distortion

MCD results are analysed at both the domain and system levels. Domain-level analyses characterise overall spectral fidelity across speech domains, while system-level analyses compare the spectral reconstruction performance of individual TTS systems within each domain.

7.4.1. Domain-Level MCD Results

Across all 960 pairs the global mean MCD was 8.98 dB ($\sigma = 2.77$ dB). Table 9 reports domain-level aggregates.

Table 9: Domain-level mean MCD (dB) aggregated across all four TTS systems ($n = 240$ pairs per domain). Lower values indicate better spectral envelope fidelity.

Domain	Mean MCD (dB)	SD
Conversational	6.90	1.76
Literary/Storytelling	7.87	1.56
Formal	9.12	1.79
Emotional	12.03	2.69

Table 7

Per-system Resemblyzer cosine similarity across four speech domains ($n = 60$ pairs per system-domain cell). Higher values indicate better speaker identity preservation. Bold denotes the best-performing system within each domain. All reference and synthetic audio is male-speaker; MMS-TTS uses a fixed single-speaker voice, so its scores reflect speaker-identity mismatch rather than gender mismatch.

Domain	Edge TTS	Gemini TTS	Indic-Parler-TTS	MMS-TTS
Conversational	0.6916	0.6303	0.7293	0.5857
Literary/Storytelling	0.7393	0.6676	0.7542	0.6000
Formal	0.5398	0.5647	0.5172	0.6814
Emotional	0.5610	0.5196	0.5241	0.5700
Overall Mean	0.6329	0.5956	0.6312	0.6093

Table 8

Cross-system mean Resemblyzer cosine similarity by emotion within the Emotional domain ($n = 10$ pairs per system per emotion). Higher values indicate better speaker identity preservation. Emotions are listed in descending order of cross-system mean similarity.

Emotion	Edge	Gemini	Indic	MMS	Mean
Sad	0.5828	0.5412	0.5375	0.5855	0.5618
Fear	0.5756	0.5343	0.5204	0.5759	0.5516
Anger	0.5662	0.5059	0.5618	0.5580	0.5480
Disgust	0.5383	0.5339	0.5155	0.5701	0.5395
Happy	0.6106	0.5103	0.4949	0.5243	0.5350
Boredom	0.5169	0.4926	0.5105	0.5931	0.5283

The Conversational domain achieves the lowest mean MCD (6.90 dB), while the Emotional domain produces the highest (12.03 dB), representing a 74.3% increase in spectral distortion. The Literary/Storytelling domain occupies an intermediate position (7.87 dB), ranking second despite containing narratively complex, stylistically marked speech, a finding elaborated in Section 8.2. The Formal domain produces higher mean MCD (9.12 dB) than Literary/Storytelling, attributable in part to within-domain heterogeneity arising from its two source subsets (FLEURS and UrduSpeech), as reflected in its elevated standard deviation (1.79 dB) relative to Literary/Storytelling (1.56 dB).

7.4.2. System-Level MCD Results

Table 10 reports MCD per system per domain. Edge TTS achieves the lowest MCD in four of five domain-subset conditions (Conversational, Literary/Storytelling, and both Formal subsets), establishing it as the most spectrally accurate system overall, with a mean MCD of 7.91 dB. MMS-TTS produces the highest overall mean MCD (9.93 dB), a difference of 2.02 dB relative to Edge TTS, and the highest MCD in the Conversational, Literary/Storytelling, and Formal (Humairawan) conditions. Within the Emotional domain, system rankings shift notably: Indic-Parler-TTS achieves the lowest MCD (10.62 dB) despite ranking second overall, while Gemini TTS, which performs competitively in other domains, produces the highest Emotional MCD (13.58 dB). This rank reversal on the most challenging domain is a substantive finding examined in Section 8.1.

7.5. Fundamental Frequency RMSE

F0 RMSE results are presented at the domain level before being disaggregated by system to examine the pitch contour accuracy of individual TTS models within each speech domain.

7.5.1. Domain-Level F0 RMSE Results

The global mean F0 RMSE across all 960 pairs was 619.34 cents ($\sigma = 229.44$ cents), with a coefficient of variation of 0.371, higher than the MCD coefficient of variation of 0.309. This indicates that pitch accuracy is more variable across the system-domain space than spectral envelope fidelity. Table 11 reports domain-level aggregates.

Table 11: Domain-level mean F0 RMSE (cents) aggregated across all four TTS systems ($n = 240$ pairs per domain). Lower values indicate better pitch contour accuracy.

Domain	Mean F0 RMSE (cents)	SD
Literary/Storytelling	521.24	89.63
Formal	520.98	95.67
Conversational	546.02	114.25
Emotional	888.54	288.75

The F0 RMSE domain ranking differs from the MCD ranking in one notable respect: Literary/Storytelling and Formal are ranked first and second respectively on F0 RMSE, whereas Conversational ranked first on MCD. This dissociation indicates that Conversational speech, while spectrally close to the reference, exhibits greater pitch trajectory deviation than Literary or Formal speech. A plausible explanation is that conversational Urdu employs more varied intonational patterns than read or narrated speech, producing larger F0 deviations even when the spectral character of the voice is well preserved. The Emotional domain again stands apart, with a mean F0 RMSE of 888.54 cents representing a 70.4% increase over the next highest domain (Conversational at 546.02 cents). The standard deviation of 288.75 cents on the Emotional domain

Table 10

Per-system mean MCD (dB) across four speech domains ($n = 60$ pairs per system-domain cell). Lower values indicate better spectral envelope fidelity. Bold denotes the best-performing system within each domain.

Domain	Edge TTS	Gemini TTS	Indic-Parler-TTS	MMS-TTS
Conversational	5.62	7.30	6.44	8.25
Literary/Storytelling	6.32	8.67	7.09	9.40
Formal (FLEURS)	7.98	11.15	10.37	10.37
Formal (UrduSpeech)	7.47	8.24	7.51	9.90
Emotional	12.17	13.58	10.62	11.75
Overall Mean	7.91	9.79	8.41	9.93

Table 12

Per-system mean F0 RMSE (cents) across four speech domains ($n = 60$ pairs per system-domain cell). Lower values indicate better pitch contour accuracy. Bold denotes the best-performing system within each domain.

Domain	Edge TTS	Gemini TTS	Indic-Parler-TTS	MMS-TTS
Conversational	500.95	555.40	592.89	535.77
Literary/Storytelling	505.53	537.21	550.12	492.08
Formal (FLEURS)	570.77	620.41	470.66	470.66
Formal (UrduSpeech)	482.63	490.04	541.34	521.31
Emotional	765.14	1027.44	915.53	846.05
Overall Mean	565.00	646.10	614.11	573.17

is more than three times that of Formal (95.67 cents), reflecting the high inter-utterance variability inherent in emotionally marked speech.

7.5.2. System-Level F0 RMSE Results

Table 12 reports F0 RMSE per system per domain. Edge TTS achieves the lowest F0 RMSE in four of five domain-subset conditions, with an overall mean of 565.00 cents. MMS-TTS ranks second overall (573.17 cents), closely trailing Edge TTS despite ranking last on MCD, indicating a dissociation between spectral and prosodic accuracy for this system. Gemini TTS produces the highest F0 RMSE in the Emotional domain (1027.44 cents), exceeding the chance-level pitch deviation expected from a system generating unconditioned prosody, and ranking last overall on this metric (646.10 cents). Within the Formal FLEURS subset, Indic-Parler-TTS and MMS-TTS produce identical F0 RMSE values (470.66 cents), suggesting that both systems converge on a similar pitch strategy for read-speech prompts from this source, likely reflecting shared training on formal read-speech data.

7.6. Cross-Metric Synthesis

The preceding subsections reported each evaluation paradigm independently. Considered jointly, however, the evaluation framework captures three complementary dimensions of TTS performance: a *holistic perceptual* quality (MUSHRA, ABX), a *spectral-acoustic fidelity* (MCD), and a *prosodic and speaker identity preservation* (F0 RMSE and Resemblyzer similarity, respectively). The central question is not which system wins on aggregate, but where these axes agree, where they diverge, and what the divergence reveals about what each method actually measures.

7.6.1. Convergence at the Domain Level

All five paradigms agree on one point: the Emotional domain is the most difficult condition in this corpus. MUSHRA scores are lowest for every system on Emotional (Table 4), MCD is highest for every system on this domain (12.03 dB at the domain level, against a corpus mean of 8.98 dB), F0 RMSE is more than 1.6 times higher than the next-worst domain, and Resemblyzer similarity is at its global minimum (0.5437). This five-way convergence is notable precisely because the paradigms differ so substantially in what they operationalize: MUSHRA reflects integrated human preference, MCD reflects frame-level spectral match, F0 RMSE reflects pitch trajectory accuracy, and Resemblyzer reflects speaker-embedding distance. Agreement across all four *independent* measurement constructs constitutes considerably stronger evidence for the Emotional domain’s difficulty than any single metric could provide in isolation. This pattern implies that affective prosody is not merely under-rated by listeners; it is genuinely harder to reproduce at the acoustic level, and the difficulty is multidimensional rather than confined to pitch, timbre, or spectral shape alone.

7.6.2. Divergence at the System Level

The system-level picture is considerably less consistent, and the disagreements are informative. Gemini TTS achieves the highest overall MUSHRA score (3.040) yet the highest (worst) overall F0 RMSE (646.10 cents) and the lowest overall Resemblyzer similarity (0.5956) among the four systems; it additionally produces the highest Emotional-domain MCD (13.58 dB) and the highest Emotional-domain F0 RMSE (1027.44 cents) of any system in this study. In other words, the system that listeners preferred most was, on every objective acoustic measure examined, the system whose output deviated most from the reference signal. This is not a small or marginal discrepancy; it is the single largest subjective-objective inversion in the dataset.

A plausible explanation, consistent with established findings in expressive speech synthesis, is that MUSHRA-style ratings capture *perceived naturalness and well-formedness*, not *reference fidelity*. Gemini TTS likely generates fluent, confidently articulated prosodic contours that sound natural in isolation even when they do not track the idiosyncratic pitch excursions or spectral irregularities of the specific reference recording – irregularities that, in the Emotional domain particularly, may themselves be partially attributable to expressive variability rather than to a canonical “correct” target. Listeners evaluating a synthesized clip against a reference and an anchor in a MUSHRA paradigm are not penalized for prosodic divergence per se; they are rewarded for sounding convincing. MCD and F0 RMSE, by contrast, penalize any departure from the reference trajectory, regardless of whether that departure sounds more or less natural to a listener. The divergence between the perceptual and signal-fidelity metrics therefore indicates that reference-matching and perceived quality are partially orthogonal constructs for emotionally expressive speech: a system can sound convincingly emotional while diverging substantially, in cents and decibels, from the specific acoustic realization a human speaker happened to produce.

A complementary disagreement appears for Indic-Parler-TTS, which ranks lowest or near-lowest on MUSHRA in three of four domains (Table 4) yet achieves the highest Resemblyzer similarity overall (0.6312) and the lowest Emotional-domain MCD (10.62 dB). Here the inversion runs in the opposite direction: a system that is objectively close to the reference, both spectrally and in speaker identity, is nonetheless rated poorly by listeners. This pattern is consistent with Indic-Parler-TTS’s description-conditioned architecture producing output that matches the coarse spectral envelope and speaker embedding of the reference while exhibiting local artifacts, monotone delivery, or unnatural microprosody that frame-level and embedding-level metrics are not designed to detect. Embedding-based similarity and cepstral distance operate at a granularity – global spectral shape, fixed-dimensional speaker representation – that does not capture the kind of moment-to-moment naturalness defects that drive human dispreference. Conversely, MUSHRA captures exactly these defects but provides no diagnostic insight into *why* a system sounds unnatural, since it returns a single integrated quality judgment.

The ABX results add a further dimension that neither MUSHRA nor the acoustic metrics provide. Even the systems involved in the ABX comparison (Edge-TTS and Gemini TTS) remain discriminable from the reference in 90.7% of trials in both domains tested, despite Gemini TTS’s relatively strong MUSHRA performance. This confirms that perceived *quality* and perceived *authenticity* are separable constructs: a synthesis can be rated as good or even excellent on a quality scale while still being reliably identifiable as non-human in a forced-choice discrimination task. This distinction is consequential for deployment contexts in which the goal is not merely high quality but indistinguishability from human speech (e.g., voice cloning, dubbing); MUSHRA alone would systematically overstate progress toward that goal.

7.6.3. Complementary Coverage Across Paradigms

Taken together, no single paradigm in this study subsumes the others. MUSHRA captures integrated human preference but conflates multiple underlying causes of (dis)satisfaction into one score and cannot localize whether a deficit originates in pitch, timbre, segmental quality, or speaker identity. ABX isolates discriminability from naturalness, a distinction MUSHRA cannot make on its own. MCD isolates segmental spectral accuracy but is insensitive

to global naturalness, as the Gemini case demonstrates. F0 RMSE isolates prosodic contour accuracy and is the most volatile metric across domains (coefficient of variation 0.371 versus 0.309 for MCD), consistent with pitch being the acoustic dimension most sensitive to expressive content and least constrained by training-data priors. Resemblyzer isolates speaker identity preservation independent of content and prosody, but, as the Indic-Parler-TTS and MMS-TTS cases illustrate, can be confounded by architectural properties (a fixed single-speaker output, in the case of MMS-TTS) unrelated to synthesis quality.

7.6.4. Domain-Specific Cross-Metric Effects

The Formal domain illustrates a further complication that aggregate domain-level statistics conceal. Its two constituent subsets, FLEURS and UrduSpeech, produce divergent system orderings on both MCD and F0 RMSE (Tables 10 and 12): Indic-Parler-TTS and MMS-TTS tie for the lowest F0 RMSE on FLEURS but Indic-Parler-TTS ranks worst on speaker similarity for the domain overall. This within-domain heterogeneity, also visible in Edge TTS’s split between UrduSpeech and FLEURS subsets in Section 7.3 (a 0.2324 similarity gap between subsets), indicates that “Formal” is not acoustically homogeneous in this corpus, and that conclusions drawn from its domain-level aggregate risk masking subset-specific effects that are themselves as large as the cross-domain effects the study set out to measure.

7.6.5. Integrated Interpretation

Collectively, these findings suggest that no evaluation paradigm employed here, taken alone, would have produced a complete or even directionally reliable account of system quality. MUSHRA alone would have crowned Gemini TTS the unambiguous winner; the objective metrics alone would have crowned Edge TTS instead; Resemblyzer alone would have favored Indic-Parler-TTS for non-Formal domains. The disagreement is not noise – it recurs consistently across the Emotional domain in particular, where every paradigm agrees the task is hard but no paradigm agrees on who handles the difficulty best. This indicates that “best TTS system” is not a domain- or metric-invariant property in this evaluation, and that any single-number leaderboard claim for Urdu TTS would be an artifact of which evaluation paradigm, and which domain, was chosen to report it.

8. Discussion

This section interprets the evaluation results presented in Section 7 and discusses their broader implications for the evaluation of modern text-to-speech (TTS) systems. The findings demonstrate that synthesis quality varies substantially across systems and speech domains, while different evaluation metrics often capture complementary aspects of performance rather than yielding consistent rankings. These observations highlight the importance of multi-metric evaluation and provide insights into the strengths and limitations of current TTS systems, with Urdu serving as a representative low-resource case study.

8.1. System-Level Findings

The cross-metric synthesis indicates that system performance in this study is paradigm-dependent rather than uniform, and this dependency is itself the most informative system-level finding. Edge TTS is the only system that performs strongly across *all* objective measures (lowest overall MCD and F0 RMSE, Tables 10 and 12) while remaining competitive, though not dominant, on

MUSHRA. This profile is consistent with a synthesis strategy that tracks the reference acoustic envelope and pitch trajectory closely, which objective metrics reward directly, while producing output that listeners judge as good but not exceptional – plausibly because conservative, closely-tracked prosody reads as competent rather than expressive.

Gemini TTS exhibits the inverse profile: weakest or near-weakest on every objective measure, yet strongest on MUSHRA (3.040 overall). This is the single largest subjective–objective inversion observed in the study. A plausible explanation is that Gemini TTS prioritizes fluent, confident delivery over strict reference tracking, a trade-off that pays off in human preference judgments but does not register as a fidelity gain on signal-level metrics, which penalize any departure from the reference trajectory regardless of whether that departure sounds more natural.

Indic-Parler-TTS occupies an unusual middle position: objectively competitive on speaker identity (highest overall Resemblyzer similarity, 0.6312) and, on the Emotional domain specifically, on spectral fidelity (lowest MCD among all systems, 10.62 dB), yet consistently the lowest-rated or near-lowest-rated system on MUSHRA in three of four domains. This indicates that its description-conditioned architecture reproduces the coarse spectral envelope and speaker embedding of the reference while exhibiting local artifacts or unnatural micro-prosody that frame-level and embedding-level metrics are not designed to detect. MMS-TTS, meanwhile, is the most domain-conditional system: it ranks highest on speaker similarity in the Formal domain, consistent with training data dominated by formal read-speech, but last on Conversational and Literary similarity and consistently worst on MCD across all domains.

Considered together, these four profiles indicate that none of the systems evaluated has converged on a strategy that simultaneously satisfies signal-fidelity and listener-preference criteria. Each makes a different, identifiable trade-off, and that trade-off – not a single quality ranking – is the more accurate description of the current state of these systems for Urdu speech synthesis.

8.2. Domain-Level Variation

All five evaluation paradigms agree that the Emotional domain is the most difficult condition in this corpus: MUSHRA scores are lowest for every system, MCD and F0 RMSE are highest for every system, and Resemblyzer similarity is at its global minimum (0.5437). This convergence across methodologically independent constructs – holistic preference, spectral distance, pitch-contour distance, and embedding distance – constitutes stronger evidence for the domain’s difficulty than any single metric could provide, and indicates that affective prosody is not merely under-rated by listeners but is genuinely harder to reproduce at the acoustic level, across multiple independent acoustic dimensions simultaneously.

The reasons for this difficulty are plausibly compounding rather than singular. Emotional reference speech itself exhibits substantially higher acoustic variability than the other three domains, evident in its larger standard deviations across every metric (e.g., F0 RMSE SD of 288.75 cents versus 95.67 cents for Formal). A noisier, more variable reference target is harder for any synthesis system to track, and is also harder for human listeners to use as a stable comparison point – a factor independently implicated in the comparatively reduced ABX discrimination rate observed for Edge TTS on this domain. The domain’s difficulty therefore appears to arise jointly from the synthesis task and the evaluation task, rather than from synthesis quality alone.

The remaining three domains show a more nuanced pattern that aggregate domain-level statistics partially obscure. Conversational speech achieves the lowest MCD of any domain but does not achieve the lowest F0 RMSE, indicating that conversational Urdu is spectrally well-matched but prosodically more variable than read or narrated speech, plausibly reflecting the more varied intonational patterns characteristic of spontaneous dialogue. Literary/Storytelling, despite its narratively complex and stylistically marked content, ranks favorably on MCD and F0 RMSE alike, and achieves the highest Resemblyzer similarity of any domain – consistent with sustained, evenly-paced narrative speech providing more stable acoustic context for both synthesis and embedding extraction than shorter or more variable utterance types.

The Formal domain illustrates a further complication: its two source subsets, FLEURS and UrduSpeech, produce divergent system orderings on both MCD and F0 RMSE, and Edge TTS shows a 0.2324 similarity gap between its UrduSpeech and FLEURS subset performance. This within-domain heterogeneity indicates that "Formal" is not acoustically homogeneous in this corpus, and that domain-level aggregates for Formal risk masking subset-specific effects that are themselves as large as the cross-domain effects the study set out to measure. This pattern reinforces the broader point that domain labels in this evaluation are useful organizing categories but should not be treated as guarantees of internal acoustic homogeneity.

8.3. Metric Agreement and Disagreement Analysis

The most consequential pattern in this study is the dissociation between what MUSHRA measures and what the objective acoustic metrics measure. Gemini TTS diverges from the reference signal more than any other system on MCD, F0 RMSE, and Resemblyzer similarity, yet is the most preferred system on MUSHRA. This divergence indicates that reference-matching and perceived quality are partially orthogonal constructs for expressive Urdu speech: a system can sound convincingly natural while diverging substantially, in cents and decibels, from the specific acoustic realization a human reference speaker happened to produce. The implication generalizes beyond this dataset – a synthesis system optimized to minimize a reference-matching loss may be optimizing for a target that is only loosely coupled to the naturalness judgments that ultimately determine perceived quality.

A complementary disagreement runs in the opposite direction for Indic-Parler-TTS, which is objectively close to the reference, both spectrally and in speaker identity, yet rated poorly by listeners. Taken together with the Gemini TTS case, these two inversions show that neither subjective nor objective evaluation alone is sufficient: MUSHRA captures naturalness defects that frame-level and embedding-level metrics cannot detect, since cepstral distance and fixed-dimensional speaker embeddings operate at a granularity that does not register local artifacts or unnatural micro-prosody; conversely, the acoustic metrics capture systematic reference deviation that an integrated MUSHRA score cannot localize or explain. Each paradigm is sensitive to defects the other is structurally unable to detect.

The ABX results add a dimension that neither MUSHRA nor the acoustic metrics provide. Even Gemini TTS, the strongest system on MUSHRA, remains discriminable from the reference in 87.5–88.9% of Emotional and Literary/Storytelling trials. This confirms that perceived *quality* and perceived *authenticity* are separable: a synthesis can be rated highly on a quality scale while remaining reliably identifiable as non-human in a forced-choice

discrimination task. A quality-only evaluation would systematically overstate progress toward indistinguishability from human speech, a distinction that matters directly for applications such as voice cloning or dubbing where authenticity, not merely quality, is the deployment criterion.

8.4. Key Findings

The key findings of this study are as follows:

- **TTS system performance is inherently multi-dimensional.** No single system achieves consistently superior performance across perceptual quality, acoustic fidelity, and speaker similarity measures, demonstrating that different evaluation criteria capture complementary aspects of synthesis quality.
- **Subjective naturalness and objective reference similarity are not always aligned.** The results show clear cases where systems preferred by listeners differ substantially from those achieving the closest acoustic match to reference speech, indicating that perceived quality extends beyond signal-level similarity.
- **Emotional speech remains the most challenging synthesis domain.** Across subjective and objective evaluations, emotional speech consistently exhibits the lowest perceptual quality, highest acoustic distortion, largest pitch variation errors, and weakest speaker similarity, highlighting the difficulty of modeling expressive prosody.
- **Modern TTS systems exhibit distinct optimization trade-offs.** Different systems demonstrate different strengths, with some favoring acoustic and speaker fidelity while others achieve higher listener preference through more natural and fluent speech generation.
- **Speech quality and speech authenticity represent different evaluation objectives.** ABX discrimination results show that highly rated synthetic speech can still be reliably distinguished from natural speech, suggesting that current systems have improved naturalness without fully achieving human-level authenticity.
- **Comprehensive TTS evaluation requires multiple complementary metrics.** Combining perceptual, discriminative, acoustic, and speaker-similarity analyses provides a more complete understanding of synthesis performance than any individual evaluation measure.

9. Conclusion

This study addressed the need for systematic and comprehensive evaluation of modern text-to-speech systems across diverse communicative domains. To this end, it introduced a domain-stratified evaluation framework covering Formal, Conversational, Literary/Storytelling, and Emotional speech, and evaluated four state-of-the-art TTS systems using complementary subjective and objective measures, including MUSHRA-style listening tests, ABX discrimination, speaker similarity analysis with Resemblyzer, and acoustic evaluations based on MCD and F0 RMSE. The framework provides a structured approach for analyzing how synthesis quality varies across different speech styles and communicative settings.

The results demonstrate that no single system consistently achieves the best performance across all evaluation dimensions and speech domains. While Gemini TTS obtains the highest overall

MUSHRA-style score (3.040), indicating stronger perceived naturalness and overall quality, Edge TTS achieves the strongest objective acoustic performance, with the lowest mean MCD (7.91 dB) and lowest mean F0 RMSE (565.00 cents). Across all evaluated systems, Emotional speech emerges as the most challenging synthesis condition, exhibiting the lowest perceptual scores, highest acoustic distortion (MCD of 12.03 dB and F0 RMSE of 888.54 cents), and lowest speaker similarity (0.5437). The ABX results further indicate that current TTS systems remain distinguishable from natural speech, with listeners correctly identifying the reference-matching condition in 90.7% of trials, highlighting remaining challenges in achieving human-level expressive synthesis.

The findings reveal several broader insights into the evaluation of modern TTS systems. First, synthesis quality is inherently multi-dimensional, and system performance depends on the evaluation criterion considered. Second, subjective naturalness and objective reference similarity are not always aligned, demonstrating that perceptual quality cannot be fully characterized by signal-level fidelity alone. Third, current TTS systems exhibit distinct optimization trade-offs, with different models prioritizing naturalness, acoustic similarity, or speaker identity. Finally, the results show that perceived quality and authenticity are separate evaluation objectives, as highly rated synthetic speech can still be distinguishable from natural speech.

Overall, this study contributes: (i) a domain-stratified benchmark for evaluating TTS systems across diverse speech conditions; (ii) a multi-metric evaluation framework that integrates perceptual, acoustic, prosodic, and speaker-similarity analyses; and (iii) a publicly released evaluation pipeline, result tables, and reproducible resources that can support future research on speech synthesis evaluation. By demonstrating the importance of domain-aware and multi-dimensional assessment, this work provides a foundation for more comprehensive and context-sensitive evaluation of TTS systems, particularly in low-resource and underrepresented language settings.

10. Future Work

Future work can extend this benchmark in several directions. First, larger and more diverse listener studies using standardized online MUSHRA platforms would further improve the reliability and comparability of subjective evaluations. Second, the framework can be expanded to include additional speech scenarios and emerging TTS systems to investigate how synthesis quality varies across different communicative contexts and technological approaches. Finally, applying the proposed evaluation framework to other low-resource and underrepresented languages would help establish more inclusive and standardized practices for speech synthesis evaluation beyond high-resource settings.

Data Set

The annotated evaluation data, including subjective listening test responses and benchmark results, is publicly available at: <https://github.com/Shifa402/Urdu-Synthetic-Speech-Evaluation>. The annotated dataset used in this study can be accessed with prior permission via this link: https://drive.google.com/file/d/1MfeXJ-LWh1SXptU5gUbXckh27YLMURNW/view?usp=drive_link

Declaration of generative AI technologies in the writing process

During the preparation of this work the authors used Claude in order to improve the readability and language of the manuscript. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

CRediT authorship contribution statement

Ali Jafar: Conceptualization, Methodology, Software, Formal Analysis, Writing – Sections 1, 2, 6 (Introduction, Related Work, Experimental Setup). **Amal Sarmad:** Subjective Evaluation Setup, Writing – Sections 5, 7, 8 (Evaluation Methodology, Results, Discussion). **Shifa Yousaf:** Literature Review, Dataset Curation, Dataset Documentation, Writing – Sections 3–4, 9–10 (Methodology for TTS System Selection, Domain-Wise Dataset Setup, Conclusion, Future Work). **Maryam Bashir:** Conceptualization, Methodology, Supervision, Review & Editing.

References

- AI4Bharat, 2024. Indic parler-TTS: A multilingual text-to-speech model for indian languages. <https://huggingface.co/ai4bharat/indic-parler-tts>. Model card and checkpoint, accessed December 2024.
- Anonymous, 2025. PashtoTTS-bench: Automated screening for low-resource non-Latin-script text-to-speech. *arXiv:2605.26978*.
- Arshad, H., et al., 2022. Exploring transfer learning for Urdu speech synthesis, in: *Proc. EurALLI*.
- Bailly, G., et al., 2025. Hot topics in speech synthesis evaluation. *arXiv preprint arXiv:2504.02738* Outcome of Dagstuhl Seminar 25032.
- Chiang, C., et al., 2024. Rethinking MUSHRA: Addressing modern challenges in text-to-speech evaluation.
- Chiang, C.Y., Lee, H.y., 2024. A study on the applicability of mushra to evaluate neural text-to-speech systems, in: *Proceedings of Interspeech*, pp. 2195–2199. doi:10.21437/Interspeech.2024-1354.
- Chung, H.W., et al., 2022. Scaling instruction-finetuned language models. *arXiv preprint arXiv:2210.11416*.
- Conneau, A., Ma, M., Khanuja, S., Zhang, Y., Axelrod, V., Dalmia, S., Riesa, J., Rivera, C., Bapna, A., 2022. FLEURS: Few-shot learning evaluation of universal representations of speech, in: *2022 IEEE Spoken Language Technology Workshop (SLT)*, pp. 798–805.
- Google DeepMind, 2025. Gemini 2.5: Our Most Intelligent AI Model. Technical Report. Google DeepMind. Accessed April 2025.
- International Telecommunication Union, 2015. Method for the subjective assessment of intermediate quality level of audio systems. Recommendation BS.1534-3. ITU-R.
- Jain, P., et al., 2022. Towards building text-to-speech systems for the next billion users. *arXiv preprint arXiv:2211.09536*.
- Kim, J., Kong, J., Son, J., 2021. Conditional variational autoencoder with adversarial learning for end-to-end text-to-speech, in: *Proceedings of International Conference on Machine Learning (ICML)*, pp. 5530–5540.
- Kirkland, A., Mehta, S., Lameris, H., Henter, G.E., Szekely, E., 2023. Stuck in the MOS pit: A critical analysis of MOS test methodology in TTS evaluation, in: *Proc. SSW*.
- Kominek, J., Black, A.W., 2008. Synthesizer voice quality of new languages calibrated with mean MCD, in: *Proc. SLTU*.
- Kong, J., Kim, J., Bae, J., 2020. Hifi-gan: Generative adversarial networks for efficient and high fidelity speech synthesis, in: *Advances in Neural Information Processing Systems (NeurIPS)*.
- Kubichek, R., 1993. Mel-cepstral distance measure for objective speech quality assessment, 125–128.
- Lacombe, Y., Srivastav, V., Gandhi, S., 2024. Parler-TTS. <https://github.com/huggingface/parler-tts>.
- Le Maguer, S., et al., 2024. The limits of the Mean Opinion Score for speech synthesis evaluation. *Computer Speech & Language* 84.
- Lyth, D., King, S., 2024. Natural language guidance of high-fidelity text-to-speech with synthetic annotations. *arXiv:2402.01912*.
- Microsoft Corporation, 2023. Azure Cognitive Services: Neural Text-to-Speech Overview. Technical Report. Microsoft Corporation. Accessed December 2024.
- Naseem, S., et al., 2025. Evaluating and developing TTS models for Urdu and Punjabi, in: *Proc. Interspeech*.
- Pratap, V., Tjandra, A., Shi, B., Tomasello, P., Babu, A., Kundu, S., Elkahky, A., Ni, Z., Vyas, A., Fazel-Zarandi, M., Baevski, A., Adi, Y., Zhang, X., Hsu, W.N., Conneau, A., Auli, M., 2023. Scaling speech technology to 1,000+ languages.
- Schatz, T., Peddinti, V., Bach, F., Jansen, A., Hermansky, H., Dupoux, E., 2013. Evaluating speech features with the ABX discriminability measure: Analysis of the classical mfc/plp pipeline, in: *Proceedings of Interspeech*, pp. 1781–1785. doi:10.21437/Interspeech.2013-564.
- Tan, X., Qin, T., Soong, F., Liu, T.Y., 2021. A survey on neural speech synthesis. *arXiv preprint arXiv:2106.15561*.
- Wan, L., Wang, Q., Papir, A., Moreno, I.L., 2018. Generalized end-to-end loss for speaker verification, in: *Proc. ICASSP*, pp. 4879–4883.
- Wang, C., Chen, S., Wu, Y., Zhang, Z., Zhou, L., Liu, S., Chen, Z., Liu, Y., Wang, H., Li, J., et al., 2023a. Neural codec language models are zero-shot text to speech synthesizers, in: *Proceedings of the International Conference on Machine Learning (ICML)*, pp. 36062–36075.
- Wang, H., et al., 2023b. Generalised zero-shot text-to-speech via voice embedding, in: *Proc. Interspeech*.
- Wang, Y., et al., 2017. Tacotron: Towards end-to-end speech synthesis, in: *Proceedings of Interspeech*, pp. 4006–4010. doi:10.21437/Interspeech.2017-1452.